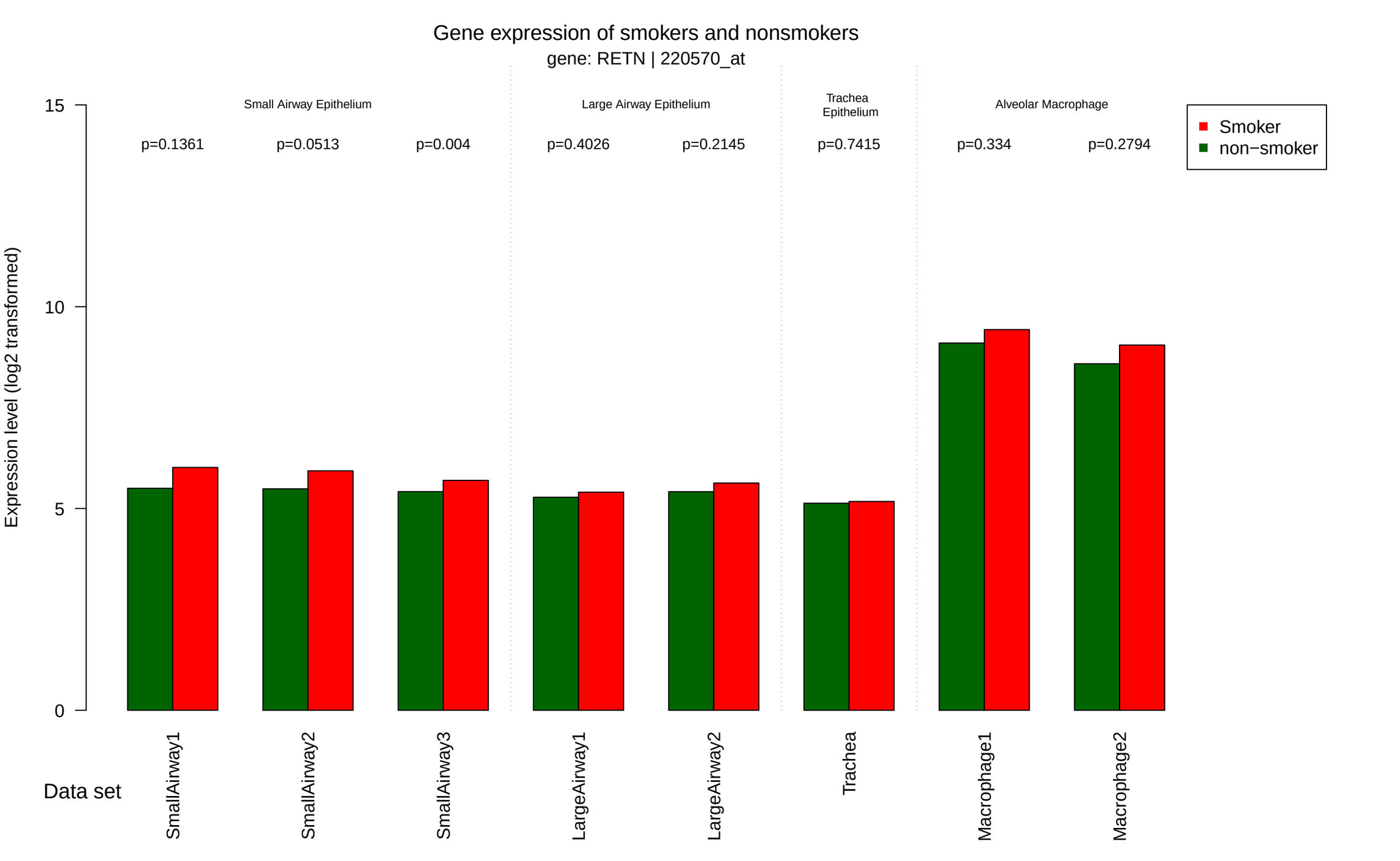
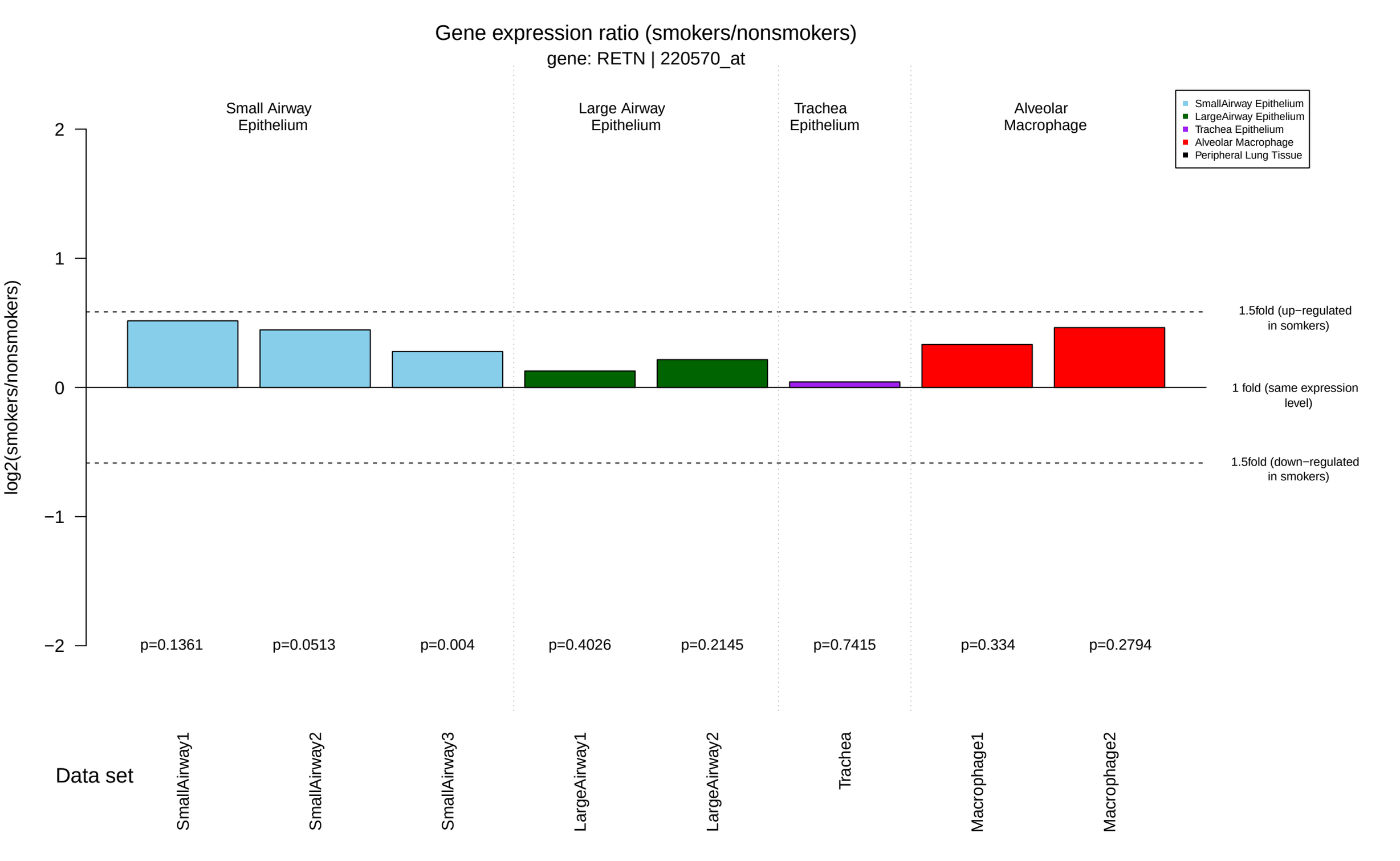
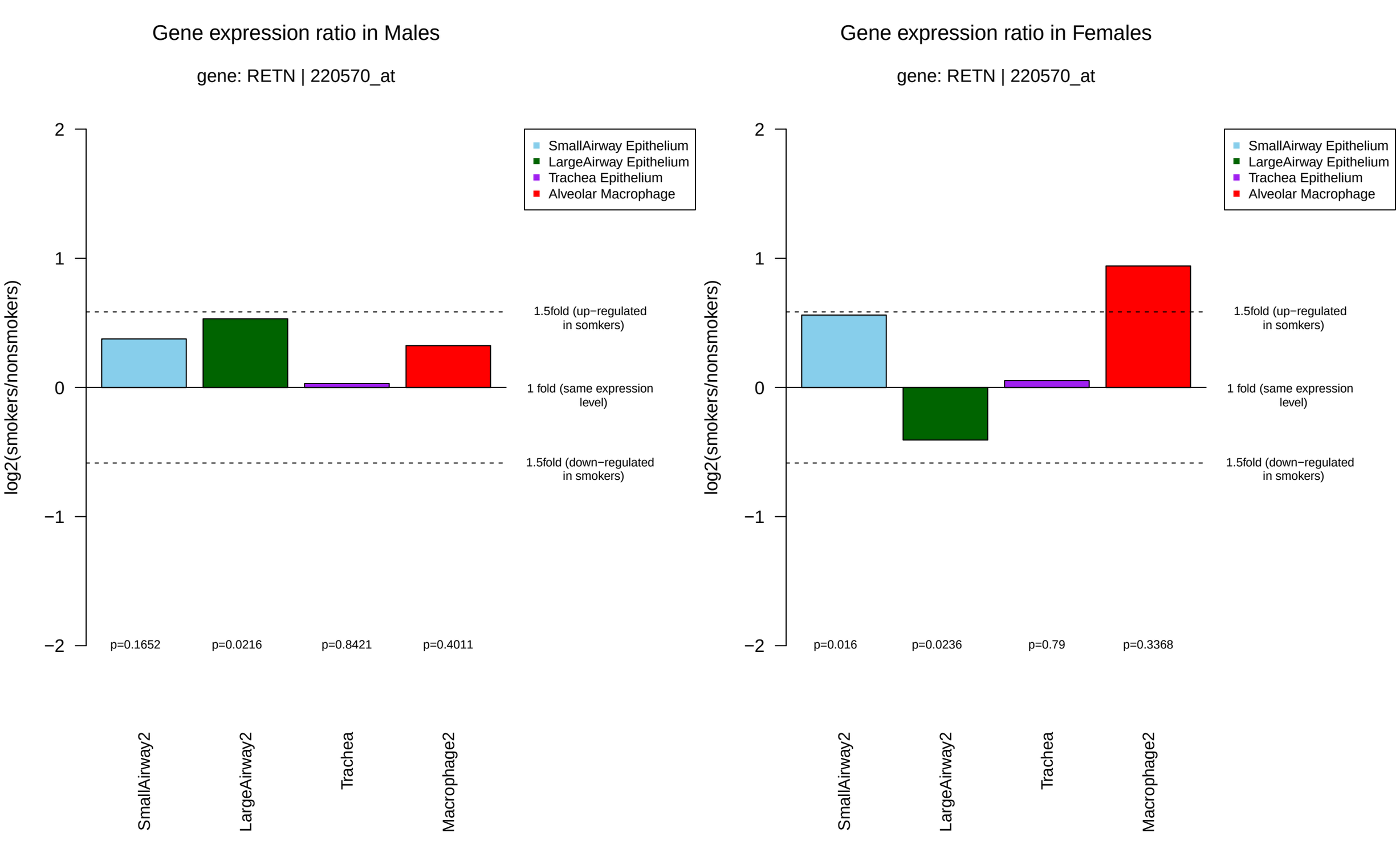
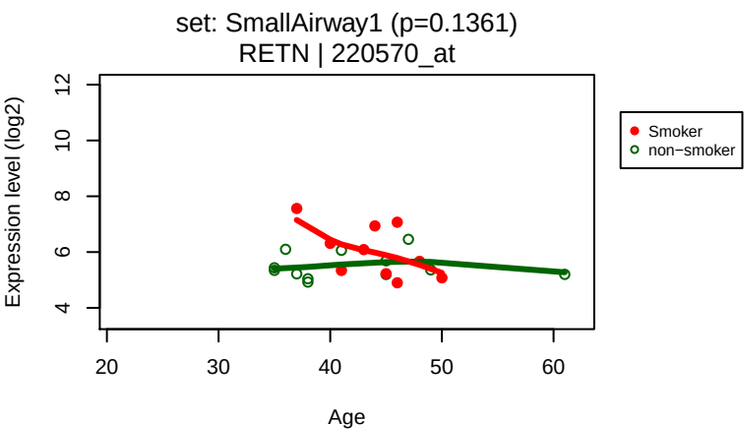
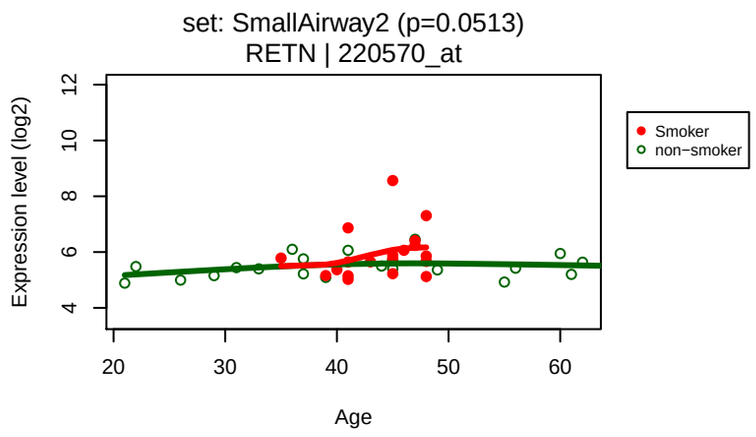
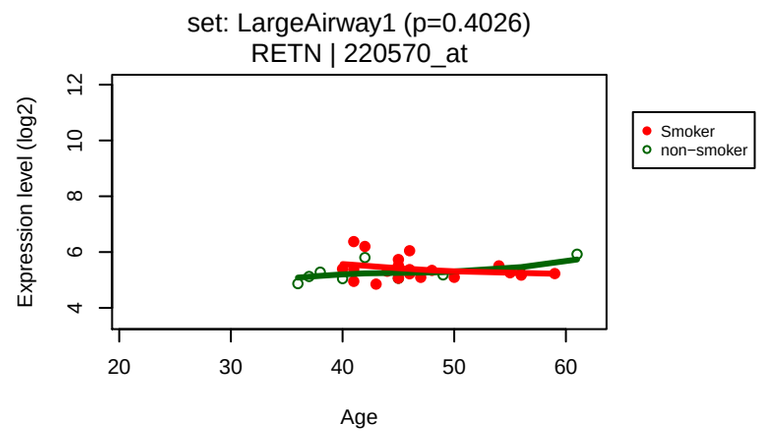

Gene expression of smokers and nonsmokers
gene: RETN | 220570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1361
p=0.0513
p=0.004
p=0.4026
p=0.2145
p=0.7415
p=0.334
p=0.2794
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

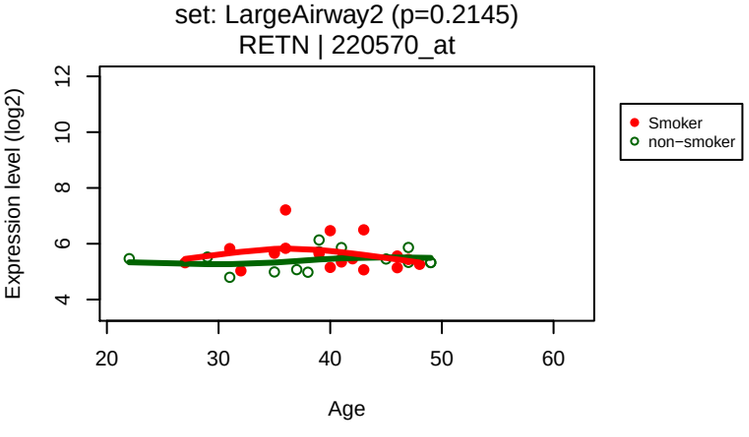

Gene expression ratio (smokers/nonsmokers)
gene: RETN | 220570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1361
p=0.0513
p=0.004
p=0.4026
p=0.2145
p=0.7415
p=0.334
p=0.2794
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

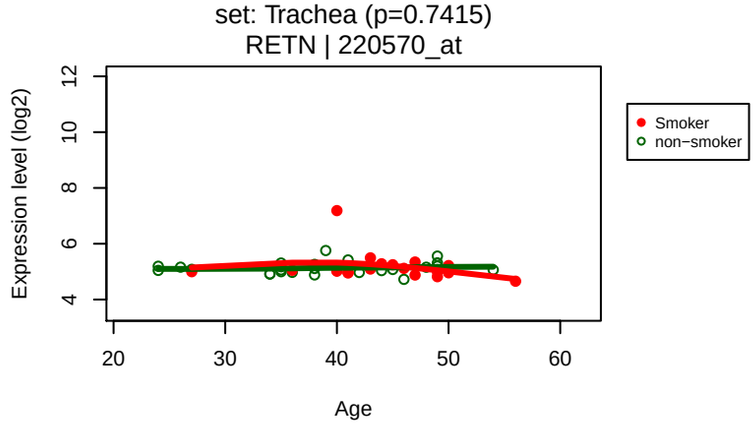

Gene expression ratio in Males
gene: RETN | 220570_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1652
p=0.0216
p=0.8421
p=0.4011
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RETN | 220570_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.016
p=0.0236
p=0.79
p=0.3368
SmallAirway2
LargeAirway2
Trachea
Macrophage2

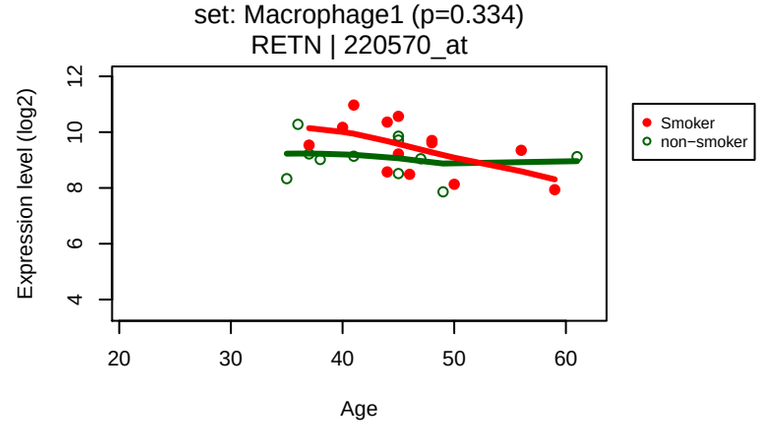

set: SmallAirway1 (p=0.1361)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker

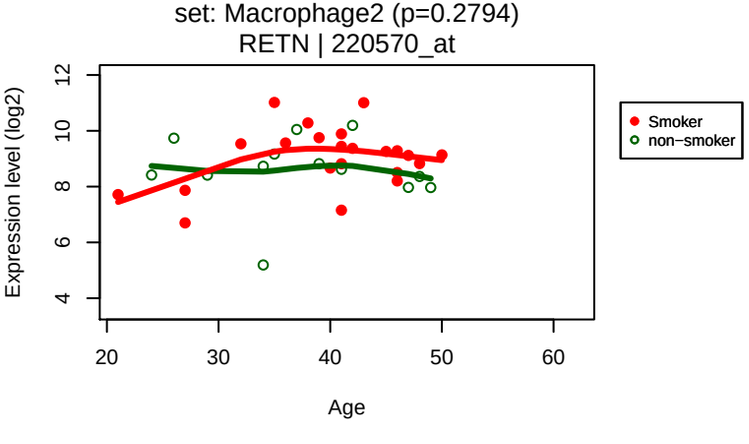

set: SmallAirway2 (p=0.0513)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker

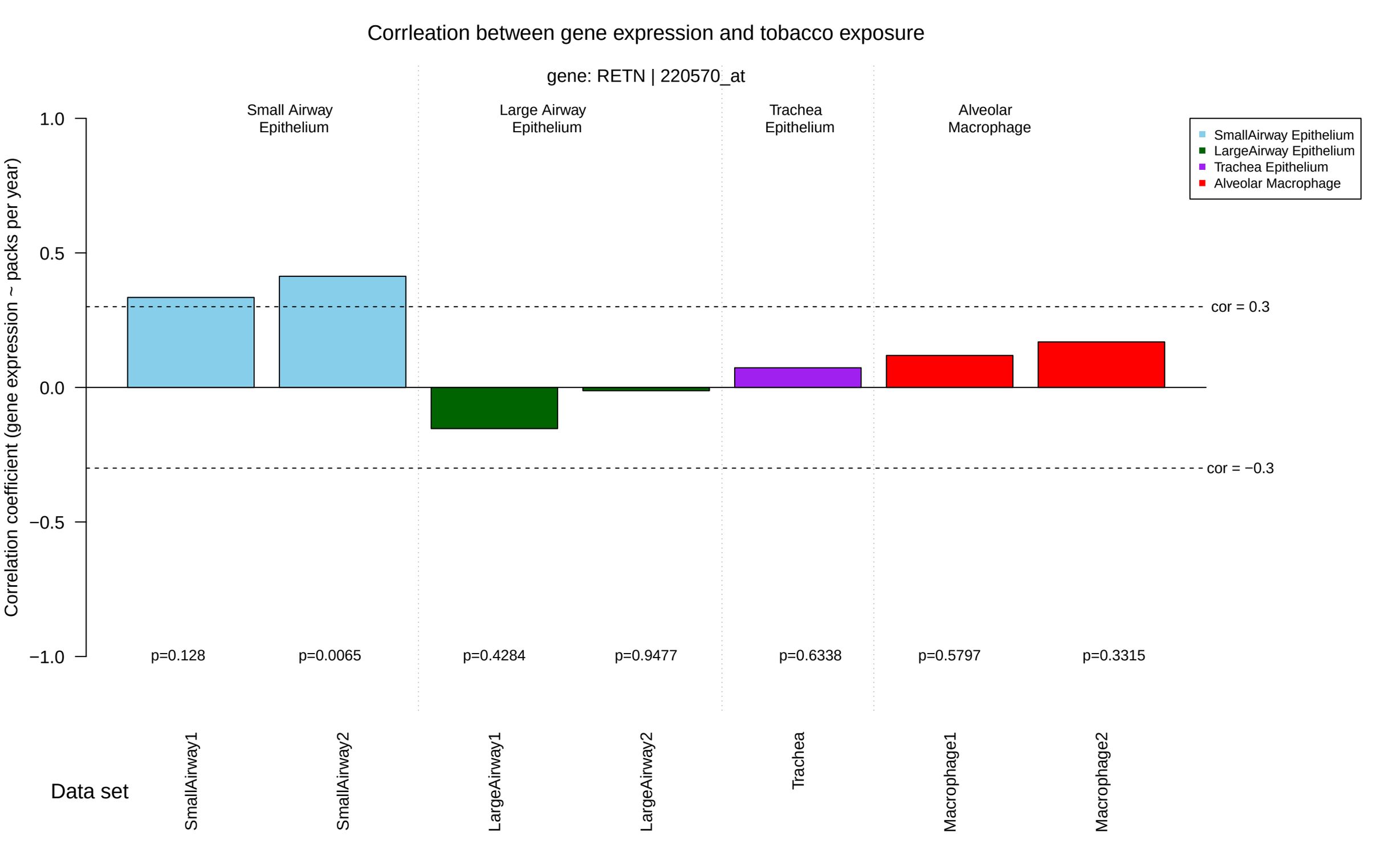

set: LargeAirway1 (p=0.4026)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2145)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7415)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.334)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2794)
RETN | 220570_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RETN | 220570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.128
p=0.0065
p=0.4284
p=0.9477
p=0.6338
p=0.5797
p=0.3315
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2